Wi-Fi Firmware - Installation Guide

Post-install wireless setup for Dell PCs running Debian-based Raspberry Pi Desktop.

Overview

The install-wifi-firmware.sh script detects many PCI/PCIe and identifiable USB Wi-Fi adapters, selects an appropriate Debian firmware package, installs it using APT, and checks for a wireless network interface. It uses the computer’s actual hardware IDs. It does not require Raspberry Pi hardware or create a Wi-Fi connection for you.

Requirements

An APT-based Debian system; Bash; sudo/root access for installation; and a temporary internet connection through Ethernet or USB tethering. Firmware availability depends on the Debian release and configured repositories.

> Before installing: Connect Ethernet or USB tethering. Without working internet access, the installer cannot download firmware packages.

1. Install firmware

```
sudo bash install-wifi-firmware.sh
```

The script prints detected Wi-Fi devices and suggested packages, refreshes APT package lists, installs available firmware, and checks whether a wireless interface is present. Reboot after installation:

```
sudo reboot
```

2. Diagnose first (no changes)

```
bash install-wifi-firmware.sh --diagnose
```

Displays adapter IDs, any wireless interfaces already found, and suggested firmware packages. This mode does not install packages or modify repositories.

3. Installer options

Command	Purpose
--diagnose	Hardware scan only; no changes.
--no-repo	Install using existing APT sources only; do not add a firmware repository.
--help	Display built-in usage instructions.

Hardware Support & Troubleshooting
What the installer recognizes, changes, and how to investigate failures.

4. Firmware matching

Packages are suggested by detected chipset vendor, not guaranteed exact-card compatibility. The script handles common Intel, Realtek, Broadcom, Atheros/Qualcomm, MediaTek, and Marvell IDs. Depending on the Debian release, it may suggest:

Vendor	Common package
Intel	firmware-iwlwifi
Realtek	firmware-realtek
Broadcom	firmware-brcm80211
Qualcomm / Atheros	firmware-atheros
MediaTek	firmware-mediatek or firmware-misc-nonfree
Marvell	firmware-libertas

5. Repository behavior

If a required firmware package is unavailable, the installer may add an official Debian non-free-firmware source on systems identified as Debian 12 (bookworm) or Debian 13 (trixie). The source is written to:

```
/etc/apt/sources.list.d/wifi-firmware-autosetup.list
```

Use --no-repo to disable this behavior. The installer does not add third-party driver repositories or install drivers from GitHub.

6. Troubleshooting

No Wi-Fi adapter detected: Check BIOS/UEFI settings and whether the adapter is present. Try lspci -nnk (internal PCI/PCIe) or lsusb (USB), if installed.
APT update or installation fails: Confirm Ethernet/tethering, check configured Debian repositories, and ensure system date/time is correct.
Firmware installed but Wi-Fi is missing: Reboot, then check the adapter and whether it is blocked:

```
ip link
rfkill list wifi
sudo dmesg | grep -iE "firmware|wifi|wlan" | tail -n 30
```

Limitations: Some Broadcom and Realtek adapters need separate kernel drivers, newer kernels, or a different firmware package. USB adapters with generic descriptions may not be auto-detected. The script does not guarantee support for every card or automatically join a wireless network.